



BSc (Hons) in **Information  
Technology Specialized in AI**  
**IT3012 : Intelligent  
Agents**

**Faculty of Computing**

## Practical 01

Year 3 · Semester 1 · 2026

Semester 2

**Objective & Lecture Mapping:** Recall that an intelligent agent acts within a continuous loop: Sensors generate percept sequences, and Actuators execute actions to maximize a Performance measure. Use this sheet to execute practical modifications and incrementally evaluate your design against Lecture 01 and 02 theory (from basic recall to analytical classification).

### Part 1: Code Analysis & PEAS Evaluation

#### Task 0 : Setting up and getting the starter Code

1. Go to the [Git Hub Repo](#). Create a Fork of this Git Repo to your Personal Github Profile. Open it using your favourite IDE and follow the below instruction to complete the practical. The base code is in the "master" brach of the repo.

#### Task 1.1: The Environment State (\_\_init\_\_)

1. (Remember) List the four components of the PEAS framework discussed in Lecture 01.

Performance Measure ,Environment ,Actuators ,Sensors

2. (Understand) Look at the `VisualGridHuntGame.__init__` method. Which specific variables in this code represent the physical "Environment (E)" state?

`self.width, self.height, self.walls, self.food_positions, self.opponents, self.agent_pos`

3. (Analyze) Based on the variables initialized (specifically `self.opponents` ), classify this baseline environment as "Single-Agent" or "Multi-Agent". Briefly justify your choice.

Multi-Agent — `self.opponents` are independent entities that each choose their own moves and can collide with the agent.

## Task 1.2: The Perception Subsystem (`get_percept`)

4. (Remember) Define what a "percept sequence" is according to Lecture 01.

The complete history of all percepts an agent has received from the start up to now (not just the current one).

5. (Understand) Which component of the PEAS framework does the `get_percept()` method represent in our code?

Sensors

6. (Evaluate) Based on the exact dictionary returned by `get_percept()` , is this environment "Fully Observable" or

**"Partially Observable"? Explain why based on what the agent can and cannot see.**

Partially Observable — the agent only sees its own position, current-cell status, opponent positions, and counters; it can't see

### Task 1.3: Action Execution (`execute_action`)

7. (Understand) When `execute_action()` deducts points for hitting a wall, which PEAS component is being actively updated?

Performance measure

8. (Evaluate) Why is it structurally important that this metric evaluates changes in the external environment state (hitting a wall) rather than the agent's internal processing effort?

Rationality is judged by real-world outcomes, not internal computation - measuring internal effort instead of environment state would let an agent "look busy" without actually achieving

## Part 2: Guided Modification & Deep Theory Mapping

Follow the implementation instructions to inject a new hazard, then answer the questions to map your code changes back to the theoretical nature of the environment.

### Step 2.1: Extending the Environment Initialization (`__init__`)

1. **Implementation Instructions:** Declare a new trap collection attribute (e.g., `self.toxic_traps = set()`) inside `__init__`. Populate it with coordinate tuples safely avoiding position `(0, 0)`, walls, and food.

9. (Understand) If you successfully add `self.toxic_traps` to the environment but intentionally **hide** this data from the agent's sensors, how does this specifically alter the environment's "Observability" classification?

It increases partial observability - A hazard now exists in the environment state that affects the agent but is completely

## Step 2.2: Updating the Perception Subsystem (`get_percept`)

1. **Implementation Instructions:** Locate `get_percept(self)` and add a new boolean sensor key (e.g., `'smells_toxin': tuple(self.agent_pos) in self.toxic_traps`) to the dictionary returned to the agent loop.

10. (Analyze) By adding `'smells_toxin'`, you expand the agent's percept sequence. Explain how this specific sensor helps the agent act with "Rationality" (maximizing expected utility).

It gives the agent the percept needed to recognize and later avoid the hazard, letting it maximize expected utility (rationality

## Step 2.3: Modifying Action Execution & Visual Rendering (`execute_action` & `draw_grid`)

**1. Implementation Instructions:** In `execute_action`, check if the updated `self.agent_pos` intersects with `self.toxic_traps` to decrement `self.score` by 15 points. In `draw_grid`, iterate through traps to render custom purple shapes on the canvas.

**11. (Remember) You programmed a severe score penalty for stepping on a trap to avoid metric exploitation. What was the classic "Vacuum World Exploit" example discussed in Lecture 01 that illustrated the danger of faulty metrics?**

A vacuum agent rewarded on its own belief/action instead of the true environment state can exploit the metric (e.g., faking

Finalize and Lock Lab 01 Analytical Evaluation



**FACULTY OF COMPUTING**